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Kenco Engineering Corporation: Strategy-Driven Costing and Lean Management

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HISTORY, CHANGE, AND ENSUING LOSS

Reading the 1988 financial statements, Ken Lutz's sons knew their company was in trouble. Their family-owned California manufacturing company had just experienced a reported loss of $350,000—a loss that was approximately one-third of the company's equity. The company is small, with $4-6 million in sales. Although it had sought business with original equipment manufacturers (OEMs), sales are primarily to custom-designed equipment end-users. Sales are obtained through bids based on the custom design characteristics of the parts Kenco manufactures.

The company is now in its second generation of family management and adheres to the same strategy initiated by Ken Lutz, the company's founder—making sales by adding value to customers' equipment.

Its foundry-castings business segment is largely outsourced for manufacturing and is not the focus of this case. Kenco's other activity, and now its largest business segment, is the manufacture of uniquely specified steel blades that are bolted to the edges of customers' heavy equipment, such as road grader original-equipment blades or earth-moving tractor buckets. Kenco's engineers work with customers and add their expertise to design the application of tungsten carbide to these add-on blades (called "bolt-ons"). Their unique tungsten carbide process hardens the edge and saves the equipment from abrasion and wear. The manufacturing process is called "tungsten carbide impregnating," or TCing.

Kenco competes based on custom-design engineering, timely delivery, and its proprietary manufacturing process. It adds value to customers' equipment, since its relatively inexpensive bolt-ons extend the life of the much more expensive OEM equipment. For example, a $3,000 Kenco bolt-on protects a $50,000 tractor bucket. Generally, it takes two to three times longer for a Kenco-TCed blade to wear out compared to competitors' replacement blades. Thus Kenco's blades lengthen the in-service time of heavy equipment, and Kenco's currently fast cycle time helps customers avoid extended downtime for difficult-to-replace blades.

Kenco has a strategic advantage with the process that melts the bolt-on blades' edges and impregnates them with tungsten carbide. Its TCing operations are in-house, and it typically processes small orders with lot sizes of 1 to 10 units. Single-unit orders are common.

Kenco was profitable from the beginning, and has always focused its efforts on its competitive advantage in designing custom products. For its TCing business, it historically paid little attention to critically evaluating and managing materials sourcing, its manufacturing processes, cost control, and job profitability. It wasn't until the 1988 financial loss that Kenco's owners were forced to deal with fundamental business changes in the nature of their business and the importance of operating issues, job profitability, and information support.

PROBLEMS UNDERLYING THE FINANCIAL LOSS

At the time of the financial loss, Kenco had problems with on-time delivery. This is an important competitive factor for Kenco since lead-time impacts downtime for customers' key equipment. Unfortunately, Kenco's lead-times were irregular and often too long.

The information system for managing inventories worsened the problem. Inventory records were less than 25% accurate. Additional delays occurred because personnel had to recount inventories and pick and tag them for delivery to specific customers.

For its TCing business, in-house manufacturing time was inconsistent and, consequently, delivery delays were common. Lead-times from sale to delivery varied from five to eight weeks. Sales would increase until lead-time reached eight weeks, and then orders dropped off. When lead-time returned to five weeks, sales would again increase.

The manufacture of TCed blades followed the de facto operating rule of "sell 1 ... make 12; somebody will buy the inventory." TCing cost control focused on purchasing materials in large quantities at low-bid, maximum-discount prices, and on reducing manufacturing setup costs by producing larger-than-needed lots. Since half of the finished blades did not turn over in 12 months, there were large inventories.

TCing was run at full capacity, with 3 shifts and 40 people. The focus was on meeting sales demand, not on fully integrated process management, job profitability, or cost control. Kenco was making so much money before 1988 that the owners did not think better management was necessary.

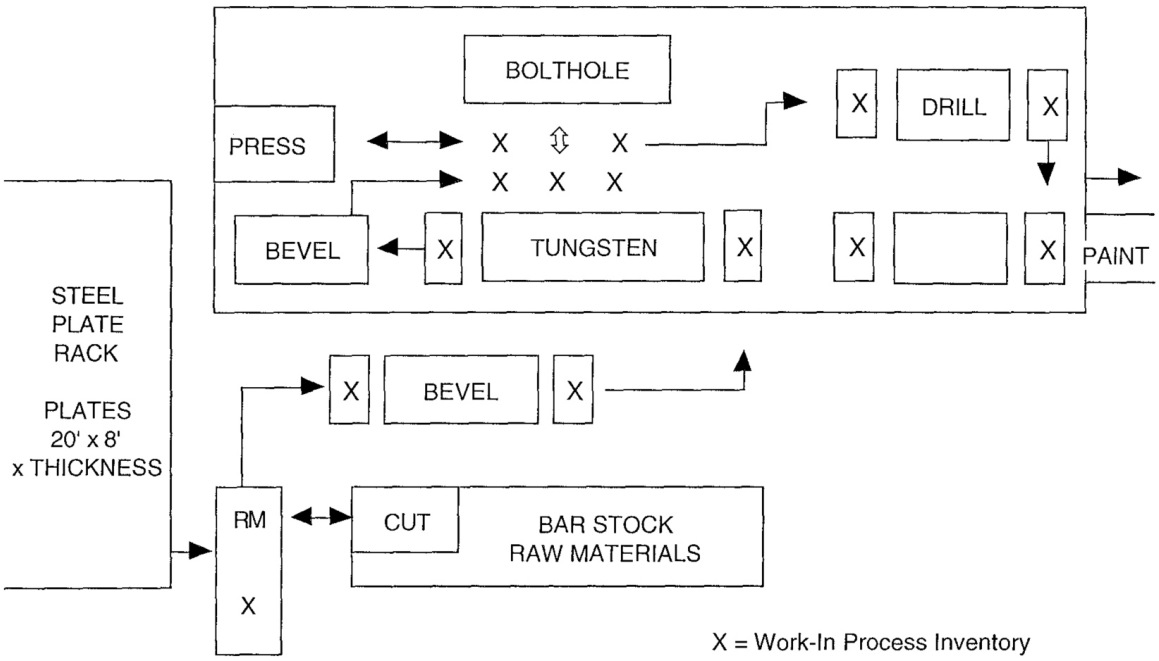
Prior to the financial loss, the practice in product pricing was to set prices at the cost of materials plus overhead and profit. Minimum bid prices for Kenco's TCing sales were targeted at 181% of expected materials cost. Thus, a casting with materials that cost $400 was sold for a minimum of $724. Managers were free to base bids on their judgment of competitive market conditions, but were expected to keep in mind the minimum-bid targets. Often, market prices greatly exceeded these minimums, and management thought all of its jobs were making a lot of money.

TC MANUFACTURING PROCESSES BEFORE THE LOSS

Before 1988, the manufacture of TCed blades was a traditional job shop with many functional workstations and multiple jobs passing through production. See Exhibit 1 for the shop-floor layout.

First, suppliers delivered large heavy steel plates to the cutting station's incoming inventory rack. "Hunt-and-peck"

Exhibit 1
Previous Production Process Layout



was required to sort through the incoming inventory and find the next job. Then an overhead crane moved the plates to workers, who cut them to rough dimensions and placed the processed materials on an outgoing rack. Forklift trucks moved the materials from cutting to beveling, and so on through the production process.

Each machine center built-up inventory on incoming and outgoing racks—as opposed to what would be required for single job flows through the workstations. The average job used 80% of its labor time in setup. Rework was high with, for example, 50% of bolt holes being drilled out-of-specification. Workers positioned materials by hand at each station, leading to crushed fingers and very high workers' compensation insurance costs.

KEY SUCCESS FACTORS

In order of importance, Kenco's key success factors are quality, on-time delivery, and cost. With quality defined as wear life and custom specifications, Kenco's products had no problems with quality. Delivery, however, was a problem. Kenco's markets are very sensitive to lead-time, and Kenco had long lead-times. With its previous systems design, Kenco was in a constant cycle of losing and gaining sales as lead-times stretched and shrunk.

An asphalt plant in Southern California, for example, discontinued Kenco's products because Kenco could not deliver on time. The failure to deliver one part idled the customer's entire plant. A part costing $2,000 idled a plant that cost $20 million! Kenco temporarily reduced lead-time by opening a second plant in Georgia. Yet when this plant's pipeline filled, the old cycle resumed. Still, Kenco produced and sold 100% of its capacity. There was no alarm until the $350,000 loss in 1988; then the cost became a problem also. Hughes would have to address these failures while maintaining quality.

THE CONTROLLER AS A CHANGE AGENT

Historically, Kenco had no controller charged with ongoing analysis of operations, job profitability, or costs. Following the 1988 loss, the Lutz brothers created a new position of controller to design and initiate the needed manufacturing, information, and analysis changes. They hired Vern Hughes, whose background included five years in public accounting and nine years in manufacturing. Hughes had in-depth knowledge of costing systems and significant large-firm experience.

The new controller developed some key ideas focusing on customer value and process simplification. Hughes would maintain the quality of custom designs, Kenco's competitive advantage in TCing, but improve on delivery strategies related to manufacturing cycle times and costs. Consistent with a belief in simplicity, Hughes thought many of the firm's practices only obscured the management process. Consequently, he sought ways to simplify management and better control costs. He also looked for ways to improve purchasing, increase control of inventory, and reduce cycle time.

NEW PURCHASING AND INVENTORY CONTROL

Hughes proposed mutually helpful linkages with Kenco's major suppliers of steel (for TCed blades). The plan was to source materials 100% from suppliers who could meet strict price, material specification, and delivery requirements. Rush orders were ended. All purchases of a given material are now at an annually negotiated price, regardless of the individual orders' unit volume. As a result, blanket discounts based on annual purchase volumes are given to Kenco, and average materials prices have dropped 15%, discounted for guaranteed sales. Suppliers delivered steel precut to job specifications, eliminating Kenco's cutting operation. All steel deliveries were in a three-day window.

Although no longer a dominant source of business, foundry casting's activity also benefited from these changes. The same materials sourcing methods used in TCing are now used for wholesaled parts, and Kenco receives four-week delivery for foundry castings.

Also regarding foundry castings, Hughes introduced cycle counting where 100 items making up 80% of sales volume are counted every four weeks. Inventory not turning over in 12 months is deeply discounted to sell, or scrapped. Wall-to-wall inventory counts are made every 90 days. Additionally, Kenco reduced 5,000 castings part numbers to 1,200 stockable items, with all other items purchased only when sold. Inventory count accuracy improved rapidly, costs were reduced, and the changes successfully rid the company of many of its management headaches.

NEW TCING CELL

Operating system simplicity, timeliness, and continuous improvement were key elements for Hughes' new system. In TCing, Hughes moved to principles of just-in-time inventory management (JIT) to improve the supply-to-delivery cycle time. Goals were to improve finished-product delivery times and simplify management of the TCing process. Kenco changed from a traditional functional layout to what Hughes calls an "in-line" (straight-line, single-unit flow) system—an approach that is similar to lean manufacturing.

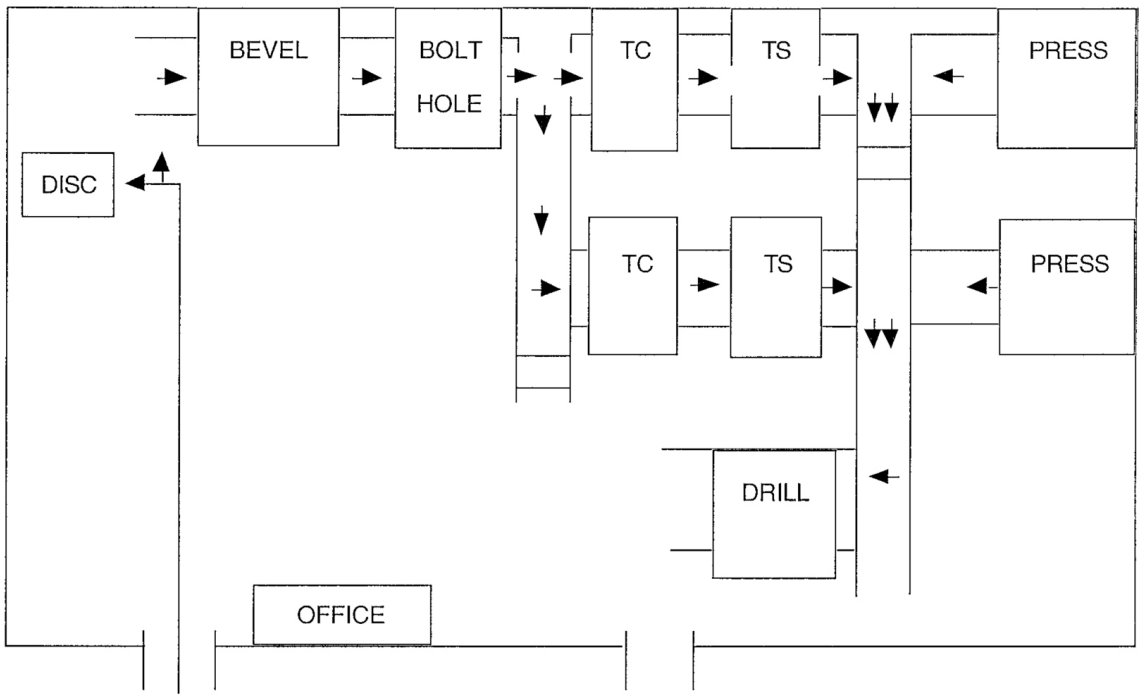
By having vendors supply steel precut to specification, which was a major dangerous activity for Kenco, the cutting operation was eliminated. Hughes regrouped the remaining production activities into two activity centers—tungsten crushing and conversion processing. Tungsten is crushed from scrap parts, and tungsten chips are merged with steel blades in a conversion processing cell. A single building houses conversion processing as one JIT manufacturing cell. See Exhibit 2 for the conversion-processing layout after the change.

The conversion-processing cell combines five of the previous manufacturing activities: (1) beveling, (2) bolt-hole cutting, (3) tungsten impregnating, (4) steel blade straightening, and (5) drilling. Process management focuses on keeping the tungsten-impregnating machine running at average available capacity, defined as maximum capacity net of average downtime for repairs and maintenance. Average available capacity works out to be 80% of maximum capacity. Thus, actual runtime below 80% signals that profits are below what is possible.

The processing flow uses conveyors for a single-unit production line through the five activities. New materials-flow equipment allows each process operator to receive and mechanically position each job. Forklifts bring steel to the beginning of the line, and remove finished product at the end. The elimination of manual positioning lessens setup time and reduces in-process inventory.

Following Theory of Constraints (TOC) principles, tungsten impregnating is deliberately designed to be the bottleneck, or capacity-limiting process. The other four activities in processing have excess capacity so that they do not limit throughput. Consequently, these design features create an unbalanced cell. Tungsten impregnating is the only constraining process, and the JIT cell serves this process.

Exhibit 2
New Conversion Processing Layout



(Precut by Supplier)

TC: Tungsten Carbide Impregnating
TS: Steel Straightening Turnaround Station

CYCLE TIME = 1 Day
SALES LEADTIME ALLOWED = 10 Days
MANUFACTURING LEADTIME ALLOWED = 8 Days

LEAN JIT CELLS

This design approach is in contrast to most lean cells that can reduce costs by striving for a balanced workflow to minimize unit costs. Instead, Hughes used an insight from the Theory of Constraints to build a cell focusing on the value of the core competency—TC. The strategy was to focus on the core value of "wear capabilities." TOC claims perfect balance is impossible and there is always a bottleneck, or constraint, that forms the focus for value creation and the center of management attention. Hughes applied the same concept of managing the constraint by deliberately designing the strategic core competency—TC as the constraint.

Kenco's cell was designed to have excess capacity in the non-TC activities to assure TC could be run at full capacity. Thus, the TC machine is the focus for all management activity, since it sets the capacity for the entire cell. In Hughes' system, changes at non-constraints were evaluation by their impact on the TC activities.

Overall, the operating system's simplicity facilitates effective process management and continuous improvement. This simplicity is a key factor underlying increased profits. Specific savings were dramatic.

1. After changes in inventory sourcing and operations, average inventory levels dropped from $80,000 to $6,000.
2. After process redesign, worker injuries were dramatically reduced, and workers' compensation insurance costs fell by more than 60%. This paid for the additional new production equipment.
3. Manufacturing labor fell 40%—from three shifts to one, and from 40 to 20 people.
4. Lead order time dropped from maximums of eight weeks to a maximum of eight days.
5. Manufacturing defects fell from 25% to one per week.

However, these were only the short-term results. Over the long term, continuous incremental improvements led to sustainable competitive advantage.

Market Prices ➔ Job Profitability ➔ Product/Process/Cost Improvement

For the TCing business, Hughes installed an information, analysis, and control system to maintain and continuously improve key areas of competitive focus and advantage. While Kenco's key success factors are product quality, on-time delivery, and cost, its continuous improvement (CI) system may be the company's ultimate competitive advantage.

With its continuous improvement system, product design and production process improvements are driven by the value of Kenco's products to its customers. This is achieved by using profitability and not just costs to evaluate jobs. At weekly production meetings, all jobs for the week are ranked by profitability, and the least profitable jobs are candidates for detailed analysis. This approach uniquely merges market value and costs to identify activities to be investigated for improvement.

This information supports a continuous improvement culture by identifying preferred activities for investigation that should have a greater chance of improving strategic value since they are the least profitable jobs. Areas for change may include product-design and manufacturing processes, pricing, and whether the product should even be continued. This information connects technological adaptations directly to market value for the products.

Conversion processing costs are allocated by time on the TCing machine. Thus, TCing time becomes the focus for cost improvement, and product redesign. By reducing a product's use of TCing machine time, it is allocated less cost and its profitability is increased. A secondary benefit of reducing a product's use of TCing machine time is that bottleneck TCing capacity is freed for other profitable sales/production opportunities.

SETTING SALES PRICES

In determining bid prices, Kenco uses prices on competing products as a guide, but negotiates a price the market will bear—a price that the customer accepts as fair. To achieve this, Dave Lutz, the president, designed a sales bidding sheet (Exhibit 3) that isolates factors expected to create market value for the product as well as providing key engineering specifications. It shows the various factors affecting production: tungsten crushing, setup, number of edges, bolt-hole cutting, etc.

Sales staff first estimate, through experience and benchmarking, the market values of activities required by a job. The values are then listed on the bidding sheet to explain pricing. The bidding sheet gives a market value rather than a cost-plus price justification. It signals unusual job specifications requiring price negotiation and communicates to customers the basis for pricing. Consequently, customers see prices as market-driven, rational, and fair. As a result of this negotiation, changes to the value of Kenco's products are readily apparent.

In application, Hughes uses these negotiated prices and the actual product cost, from his costing system, to compute the gross profit for each job to initiate continuous improvement management in weekly meetings. This system and not individual insight drives CI.

Exhibit 3
Bidding Sheet – Front End Loader Edges (FEL) A-36

Customer:		Date:	10/2/9
Dist.		Calc By:	RHM

Remarks:			
Item No. 1		DWG NO.	N/A

Type (DE/SE)	SE
Thick	1.00
Width	12.00
Length	53.00
Hole Qty	12
TC Passes	4
Single Passes	0
SU	$17.00
SQ in TC	0.00
List Price	$407.19
Dist Price	$325.75

Set Up Chrg	Single Edge	Double Edg
0"-35"	$10.00	$12.00
36"-71"	$15.00	$17.00
72"-107"	$20.00	$22.00
108"-131"	$22.00	$24.00
132"-155"	$24.00	$26.00
156"-up	$30.00	$32.00

Sq in TC	0.00
Weight	178.08
Pin TC	212
SPINTC	0

SET UP	STEEL	TC	Single TC	BOLTS	Sq in TC
$17.00	$106.85	$225.14	$0.00	$42.00	$0.00

Remarks:			
Item No. 2		DWG NO.	N/A

Type (DE/SE)	SE
Thick	1.00
Width	12.00
Length	53.00
Hole Qty	12
TC Passes	4
Single Passes	0
SU	$17.00
SQ in TC	0.00
List Price	$407.19
Dist Price	$325.75

Set Up Chrg	Single Edge	Double Edg
0"-35"	$10.00	$12.00
36"-71"	$15.00	$17.00
72"-107"	$20.00	$22.00
108"-131"	$22.00	$24.00
132"-155"	$24.00	$26.00
156"-up	$30.00	$32.00

Sq in TC	0.00
Weight	72.24
Pin TC	86
SPINTC	0

SET UP	STEEL	TC	Single TC	BOLTS	Sq in TC
$12.00	$39.73	$91.33	$0.00	$14.00	$0.00

For example, one product initially required a series of tungsten strips. Impregnating these strips as specified onto the steel blade used several passes on the TCing machine. With costs assigned using time on the TC machine, this created a reported loss on this job. Looking to future contracts for this product, Kenco worked with the customer to redesign the product. In the future, it would use fewer tungsten strips that were more critically placed. The product then became profitable, and the price remained competitive.

THE COSTING SYSTEM

The "four-wall" costing system designed by Hughes allocates all JIT cell costs based on products' consumption of TCing machine time. The allocation rate equals total costs in the JIT conversion cell divided by average available capacity. This cost assignment reflects "cause and effect" at a strategic level. While cell costs are operationally largely fixed, they are not fixed strategically. These costs are determined by the strategic choices about how to provide TCing capacity and service levels to customers. There are several advantages of this costing system.

1. It keeps the primary cost-control focus on Kenco's strategically determined value-adding capacity, which is the use of its proprietary TCing machine.
2. It is consistent with the production system's design that emphasizes a "value stream" approach to managing cell throughput with a single capacity constraint. This design is different from a traditional functional layout where individual processes within a production line are managed for efficiency, or targeted cost control. That is, the emphasis shifts to optimizing a value stream and not individual components within the system.
3. It is simple and easily understood, thus facilitating the timely weekly production and analysis of cost reports and job profitability.

Kenco's measurement of job profit equals negotiated market price minus materials cost and allocated cell costs. This profit measure has several information features.

1. It captures the net benefit Kenco receives from using its key competitive TCing capacity on a particular job, aligning strategy with operations.
2. It captures the value set by the market for Kenco's premium product.
3. Jobs with the lowest profit margins are the starting point for weekly continuous improvement activities.

Finally, if continuous improvement efforts fail through product redesign, process improvement, and rebidding the price, Kenco will simply drop a product. Kenco's owners have made a strategic decision to produce premium, high-value products and to maintain this product image, pricing structure, and market niche.

If low-profit products are turned away, this is an acceptable effect of maintaining Kenco's product and market focus. The Lutz brothers prefer to protect their long-term value, even if this means not maximizing short-run profits. As a side benefit, any idle capacity gives employees "time to tinker" and supports the continuous improvement activities that have lead the company to its dominant position.

TIMELY CONTINUOUS IMPROVEMENT PERFORMANCE ANALYSIS

Kenco stresses timely continuous improvement driven by market prices. By using profitability rather than the costing system alone to drive CI, the value of products to the customer drives specific continuous improvement efforts. Timely weekly reports rank jobs by profitability. The least profitable are investigated first for continuous improvement efforts. Importantly, the job-cost buildup is simple and easily understood by Kenco's employees and facilitates analysis.

Exhibit 4 shows the weekly production and cost report used with this analysis. It includes Kenco's weekly cost buildup per job. Note the simplicity of this buildup and its culmination in a gross profit ratio. The ratio, as discussed before, is used to maintain weekly accountability for each job, producing a threshold level of profitability. The exhibit's structure is explained below using the line 2 job data.

column 4: date job processed			9-26-1995
column 5: unit price			$13.20
column 7: cost of steel used		$2.60	
column 6: inches of tungsten carbide used	5		
cost per inch	$.17182		
column 9: cost of tungsten carbide		0.86	
column 8: conversion processing time used	.097		
cost per unit of time (station D rate)	$76.72		
column 10: cost of conversion processing time used		7.44	
column 11: total cost per unit of product			10.90
column 12: unit profit			$2.30
column 13: gross profit ratio			17.41% (insignificant rounding error as shown)

Exhibit 4
Weekly Production and Cost Report

Org. Qty.	Qty. Good	Sta #	Date	Unit Price	T.C. In.	Steel Cost	Avg. Time	T.C. Cost	Conv. Cost	Unit Cost	Unit Profit	Gross Profit
17	17	3	09-28-1995	$27.50	23	$0.00	0.202	$3.95	$34.50	$38.45	($10.95)	-39.81%
48	48	D	09-26-1995	$13.20	5	$2.60	0.097	$0.86	$7.44	$10.90	$2.30	17.41%
2	1	D	09-25-1995	$247.00	90	$86.39	0.699	$15.46	$53.63	$155.48	$91.52	37.05%
2	2	D	09-25-1995	$247.00	90	$86.39	0.654	$15.46	$50.18	$152.03	$94.97	38.45%
8	8	3	09-27-1995	$46.58	8	$17.58	0.049	$1.37	$8.37	$27.32	$19.26	41.34%
80	68	3	09-26-1995	$27.81	10	$5.44	0.048	$1.72	$8.20	$15.36	$12.45	44.78%
2	2	4	09-26-1995	$752.73	363	$95.73	1.375	$62.37	$234.82	$392.92	$359.81	47.80%
4	4	4	09-26-1995	$411.56	231	$101.22	0.42	$39.69	$71.73	$212.63	$198.93	48.34%
80	80	4	09-28-1995	$27.81	10	$5.40	0.042	$1.72	$7.17	$14.29	$13.52	48.61%
1	1	4	09-26-1995	$1,390.79	655	$384.94	1.144	$112.54	$195.37	$692.85	$697.94	50.18%
64	34	4	09-28-1995	$50.04	23	$3.45	0.102	$3.95	$17.42	$24.82	$25.22	50.40%
80	80	4	09-27-1995	$20.36	8	$4.34	0.021	$1.37	$3.59	$9.30	$11.06	54.32%
80	40	D	09-29-1995	$38.88	16	$6.82	0.101	$2.75	$7.75	$17.32	$21.56	55.46%
64	64	D	09-27-1995	$35.03	10	$6.50	0.096	$1.72	$7.37	$15.58	$19.45	55.51%
1	1	4	09-27-1995	$3,697.00	1388	$905.48	2.887	$238.49	$493.03	$1,637.00	$2,060.00	55.72%
4	4	D	09-25-1995	$205.18	47	$52.92	0.341	$8.08	$26.16	$87.16	$118.02	57.52%
2	2	4	09-25-1995	$226.74	119	$31.57	0.251	$20.45	$42.86	$94.88	$131.86	58.15%
80	40	D	09-27-1995	$27.04	10	$5.44	0.05	$1.72	$3.84	$10.99	$16.05	59.34%
1	1	4	09-25-1995	$3,328.26	1248	$809.38	1.88	$214.43	$321.06	$1,344.87	$1,983.39	59.59%
4	4	4	09-26-1995	$502.42	242	$88.26	0.422	$41.58	$72.07	$201.91	$300.51	59.81%
2	2	4	09-27-1995	$560.20	160	$49.05	0.768	$27.49	$131.16	$207.69	$352.51	62.93%
1	1	4	09-26-1995	$281.31	139	$35.30	0.257	$23.88	$43.89	$103.07	$178.24	63.36%
2	1	4	09-25-1995	$336.42	170	$50.03	0.256	$29.21	$43.72	$122.96	$213.46	63.45%
4	4	4	09-25-1995	$280.83	155	$17.68	0.317	$26.63	$54.14	$98.45	$182.38	64.94%
1	1	4	09-25-1995	$715.52	361	$92.34	0.525	$62.03	$89.66	$244.02	$471.50	65.90%
40	40	3	09-28-1995	$23.09	6	$3.41	0.017	$1.03	$2.90	$7.34	$15.75	68.19%
4	4	4	09-26-1995	$143.34	70	$11.88	0.109	$12.03	$18.61	$42.52	$100.82	70.33%

Run Time	Total Sales	Total TC Inches	Total TC Cost	Total Steel Ext.	Total Labor/OH	Total Cost	Total Profit	Gross Profit
86.304	$50,798.68	20082	$3,450.52	$10,036.95	$10,617.67	$24,129.31	$26,669.37	52.50%
	$50,798.68	20082	$3,450.52	$10,036.95	$7,488.00	$20,975.47	$29,823.31	58.71%

THE CONTINUOUS IMPROVEMENT SYSTEM AND STRATEGY

Virtually all formal continuous improvement systems have four sequential steps.

1. Activities for improvement must be selected.
2. Root causes for the activity performance as it exists must be determined.
3. Modifications must be discovered and implemented.
4. The impact of the change must be assessed.

Kenco's system follows the same pattern, with two advantages. First, the search process in step one is politically neutral. The activities to be investigated are a result of accounting calculations and not personal judgment. Second, the system provides a timely unambiguous assessment of the appropriateness of any changes. This allows timely assessment using politically neutral criteria of profitability.

Kenco's system is process-based and apolitical, focusing attention on the TCing machine, the strategically designated constraint. This is where change will be most beneficial to its strategic plan. It is, by strategic design, the only constraint.

CONTINUOUS IMPROVEMENT CONTROL CHARTS

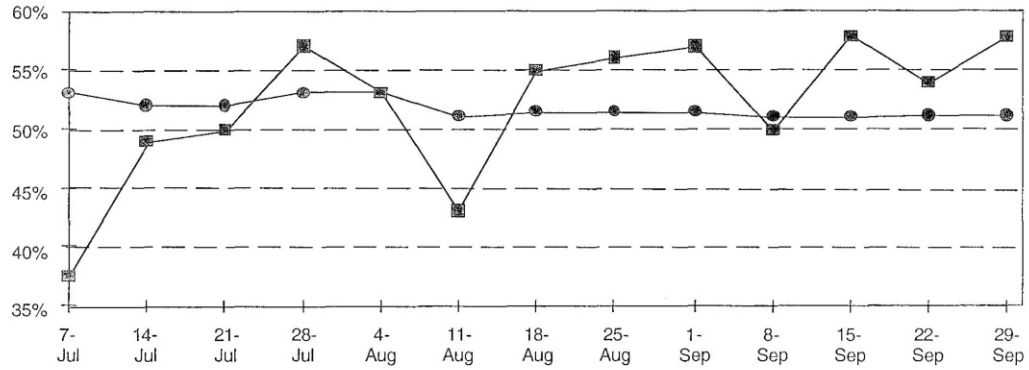
The control charts used by Kenco uniquely reflect this continuous improvement culture. Several weekly measures are compiled to track performance trends (Exhibit 5). These are not merely the deviation from budget projections, as in traditional budgeting systems, but rather focus on continuous improvement. Measures include gross profit percentages, customer service (targeted as product delivery within eight days), and tungsten-steel inches impregnated. These measures are highly correlated with the critical success factors of quality, delivery, and cost.

They are used to evaluate the continuous improvement activities. Each chart reflects trends over time using a 13-week moving average. Unlike traditional budgeting standards, improvements are "unknown and unknowable." Thus, the trend demonstrates the success or failure of CI management over time.

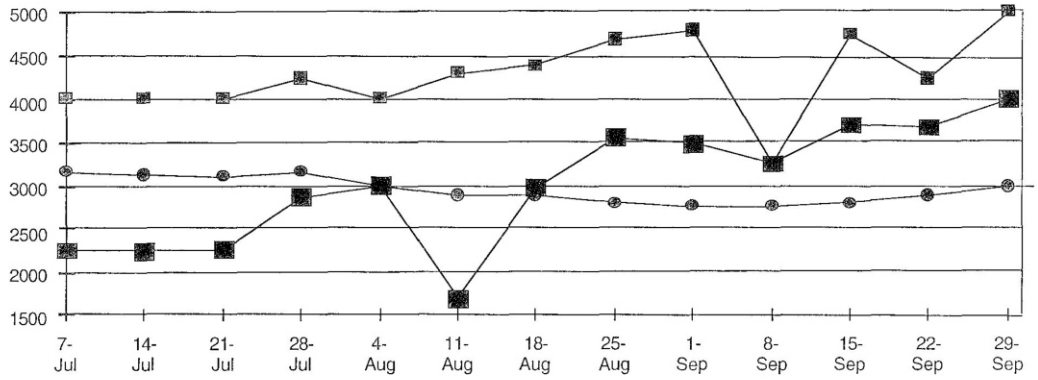
The first chart is the "Gross Profit per Week" out of the manufacturing cell and does not include administrative overhead costs. The data comes directly from the accounting report and tracks a cumulative 13-week moving average

Exhibit 5
Weekly Performance Measures

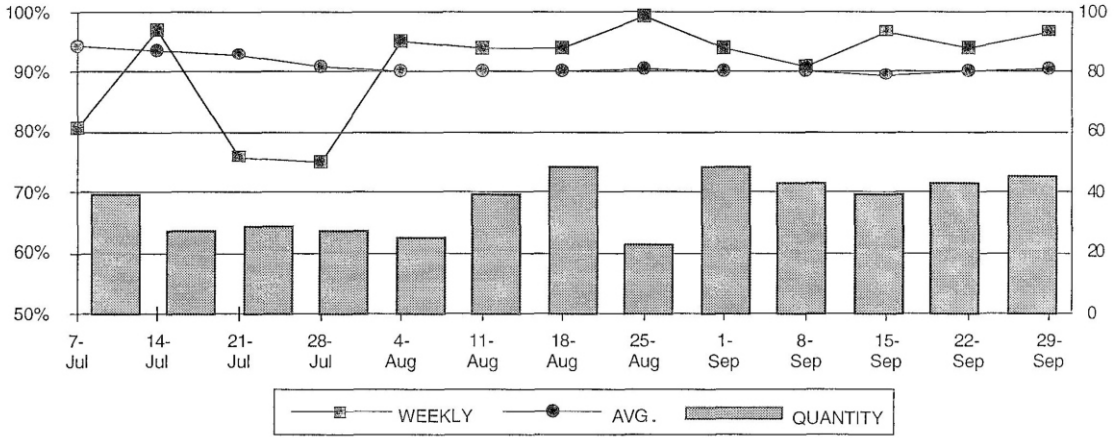
Gross Profit Per Week



T.C. Inches Per Day



Customer Service



(circles). This chart speaks to the overall progress of continuous improvement efforts reflected in profitability. This is a combination of the costs and market value. The left-hand side on the vertical axis lists the profit percentages. The horizontal axis just sets the time line.

The second graph refers to factory operations. This particular example was from the time the company had a second facility in Georgia. The squares are the production for each plant and the circles are the 13-week cumulative average TC output. Vertical axis is the number of TC inches and the time line is along the bottom. The graph speaks to customer service. The bars are the percentage of orders completed on time and the squares are weekly orders shipped. The circle is again a 13-week moving average.

All of these have the same goal of demonstrating how the company is progressing and improving. This is a distinctly different application than in a traditional organization where the goal is to meet a budget with stable standards. In a continuous improvement organization, the goal is not to meet current productivity norms but gradually to improve critical competencies over time. Thus, trend lines and charts are much more important in the management and control of continuous improvement processes.

EPILOGUE

Through a process of gradual incremental change and timely adaptation to market needs, since 1989 the company has thrived. The manufacturing process has become highly automated with computer-controlled machines. The company first opened and then closed its Georgia plant because of the productivity improvements in its home base Roseville, California, plant. The costing systems first led Kenco to outsource steel cutting and then to bring the process back into the company after more efficient processes were developed.

Products and product lines have been continuously adapted to customer needs, with current SKU's for over 1,200 specifications. While the production facility was designed for small lot sizes, shop floor personnel were able to reorganize to accommodate large orders (such as from the California Transportation Department) on a regular basis without additional input from management.

Continuous incremental change, which Kenco employees call "tinkering," is an accepted part of the CI culture. The president, Dave Lutz, commented to Vern Hughes, "If you had suggested all the changes we have made when you joined the company I would have thrown you out the door on your ear!" Hughes responded, "I had no idea it would lead to this."

Essentially the same strategic goals and accounting system have remained in place since 1989. The accounting system supports the alignment of core competitive design characteristics. Kenco's continuous improvement system, aligned with a successful strategy, has incrementally moved the company into a dominant position in its industrial niche.

SUGGESTED QUESTIONS

Q1. Is either activity-based costing (ABC) or Kenco's costing system preferable under GAAP?

Q2. Kenco's organizational design borrows from both lean management and the Theory of Constraints. Describe the unique characteristics of Kenco's design and accounting system with respect to lean management and TOC. In particular, discuss differences in approaching continuous improvement, cell design, value- versus cost-based decisions, and accounting methods.

Q3. In Exhibit 2's bolt-hole process, a bolt-hole machine drills holes into the steel blades that are to be TCed. Some blades have more and some have fewer bolt holes, depending on the design specifications. (a) Would the use of activity-based costing (ABC) to allocate bolt-hole-processing cost versus Kenco's use of time on the TCing machine lead to different product costs? (b) In what ways might the use of each of these allocation systems be beneficial or ineffective—in general, and at Kenco? Explain fully.

Q4. How do Kenco's manufacturing and reporting systems support its key success factors (key strategic components) of product quality, on-time delivery, and cost?

Q5. Describe Kenco's CI system and compare this to process change using traditional budgeting.

Q6. How would cost-plus pricing change continuous improvement at Kenco? Compare and explain fully the strategic and operational implications of cost-plus pricing versus Kenco's existing pricing approach.

Q7. Compare how the strategy for lean manufacturing and Kenco's system lead to different production and information systems. Include in your discussion the strategy underlying each approach, the information/accounting systems used to support the strategy, and decision effects.

Q8. How are Kenco's performance charts specifically designed to support Kenco's strategy and continuous improvement management process?

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